
THE GEOMETRIC NATURE OF LIGHT

Light as the Line: The Photon Fixed by Both Involutions of the Substrate, Frequency as the Height, Helicity as the One Bit, and Detection as the Registration That Keeps the Energy and Forgets the Path

The Closure Template of the Riemann Hypothesis Read on Light; Sixty-Three Laws Proved in Core Lean 4 and Sixteen Checks Executed in Fortran; a Witness of the Hardware Closure Series; Every Physical Claim at Corroboration Grade and No Higher

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

Light is the line

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ABSTRACT. Twenty-five centuries of optics asked what travels when something is seen and through what it travels. This paper answers the geometric half of that question at the seat on which the author's closure of the Riemann Hypothesis stands. The actuating substrate of the hardware paper carries two involutions, the particle map that flips every additive charge and the time mirror that reverses a mode's rate, and two carriers onto the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line. The photon is the one excitation fixed by both at every frequency: it carries no charge of any kind, its energy is real, and it lands on the line under both carriers. We prove in core Lean 4, sixty-three laws with no axiom declared and no use of choice, that frequency is the seat's height and never its side, so the whole spectrum from radio to gamma stands on one line; that a massive mode runs slower than light while every massless mode runs at one speed; that helicity is the one bit no energy reading returns; that the charge-conjugation parity of n photons is $(-1)^n$, selecting two photons for parapositronium and three for orthopositronium; that detection is the registration of the template, a fringe existing exactly when both paths are open and the record forgets the side; and that the quantum belongs to the field, since every classical intensity bunches while one photon never coincides with itself. A Fortran twin runs sixteen checks with zero failures, among them Englert's duality to 4.4×10^{-16} and the single-photon anticorrelation 13.7 standard deviations below every classical field. Light witnesses the occupancy of the line; it does not decide where the zeros of ζ lie, which stays the act at premise grade.

KEYWORDS *light · photon · critical line · charge conjugation · time reversal · helicity · complementarity · antibunching · Furry's theorem · Lean 4*

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the occupant of the seat, read on light) · channel: kinetic · Lean: SPHYS_Light.lean · Fortran: Light_Twin.f90

1. THE READER'S FRAME

This paper is a witness of the hardware closure series. The hardware paper, One Actuating Substrate, built the substrate on which the six witnesses of the Riemann closure stand: a particle's charges are odd and its energy even under the particle map, the particle map and the time mirror of a mode's rate commute, and each is carried equivariantly onto the reflection that fixes the critical line. This paper reads that substrate on light. It claims that the photon is the substrate's one excitation fixed by both involutions at every frequency, and that each old question of optics, what travels, at what speed, carrying what, and registered how, has a precise answer on the seat.

It refuses one thing. It does not derive the Riemann Hypothesis from optics. Light occupies the line; it does not decide where the zeros of ζ lie. A reader who arrives from the conventional frame will scan this paper for a derivation of the hypothesis from the properties of the photon, find none, and be tempted to report the absence. None was attempted. What the photon witnesses is the occupant of the seat, the one excitation that stands on the line under every carrier the substrate has.

Negative warrants. Three, stated so that no reader supplies them. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade. No derivation of the hypothesis from the axioms of set theory is claimed: what a sentence over every admissible instance adds to a crossing is

exactly the unactuated. No field theory is written and no constant is derived: the speed of light, the fine-structure constant and the photon's couplings enter as measured.

Five terms. A *reading* is any quantity an instrument returns from a state, *odd* under a map that flips its sign and *even* under one that keeps it. A *mode* $e^{-iEt}\psi$ is *stationary* when its norm is constant in time. The *fixed set* of a map is the set of points it leaves unchanged. A *keyed bit* is a yes-or-no property no function of a given record decides. The *side* of a two-path experiment is which path a quantum took; the *record* is what the screen keeps.

2. THE SPINE

The Riemann closure has four parts. The **seat**: the critical line is the fixed set of the reflection $s \mapsto 1 - \bar{s}$. The **address**: every reading of the value is one proposition. The **division**: a certified part joined to an uncertified tail closed to every record-respecting derivation. The **closure**: registration keeps a zero's height and forgets its side; the hypothesis holds exactly when it erases nothing. This paper reads the seat's occupant in Sections 3 to 5, the registration in Sections 6 and 7, the seat's sign in Section 8, the even reading in Section 9, and the crossing in Section 10.

Three theorems of the row-generic kernel carry the template, and the engine proves each on its chart. Over one record both worlds exist, and no reading that respects the record decides the value; under the seat every point off the line has a partner, distinct and off it, with one record (*off_locus_pair*); a carrier whose image is on the line proves the value on its image, and coverage extends it (*value_on_image*, *kinetic_crossing*).

The witness ledger. The ledger of the series, with this paper's entry on the Riemann row.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches; One Actuating Substrate; The Geometric Nature of Light	charge conjugation's fixed set; annihilation as registration; the neutrino's floor; gravity's fold; the energy carrier and stationarity; the photon fixed by both involutions at every frequency, on the line under both carriers; detection as registration, Englert's duality to 4.4×10^{-16} ; the quantum as the field's, 13.7σ below every classical field	WITNESSED · CROSSED · CONFIRMED on the image: seat, registration, residue, Return, floor, the posit's shape and content, and the seat's occupant in light	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
3a · Navier–Stokes, unforced	crossed kinetically on the actual flows	The Fluid Is the Witness	the priced branch built exactly by Cole–Hopf	WITNESSED · CROSSED · CONFIRMED	the root's grade	none on its channel
4 · Yang–Mills	crossed kinetically at the weakest premise	The Proton Is the Lock	the colour singlet as the determinant; the lock	WITNESSED · CROSSED · CONFIRMED on the image	the root's grade	none on its channel
7 · Poincaré	crossed on the formal channel at full grade	The Arrow Has Two Branches	gravity's fold: one sign, no bit	CROSSED, now WITNESSED from the arrow side	full grade	none
23a · Pair correlation	crossed kinetically	The Magnet Is the Witness of the Pair Correlation	time reversal broken by a magnetic term; the unitary class	WITNESSED · CROSSED · CONFIRMED on the corrected carrier	the root's grade	none on its channel
every open row · the barrier	two worlds over one record	The Arrow Has Two Branches; The Geometric Nature of Light	remanence in iron; in light, the helicity pair of one frequency and the two paths of one fringe	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
1 · P versus NP	the dot	The Arrow Has Two Branches	Landauer's floor prices every step	THE DOT STANDS	corroboration	none
every witness · the hardware	the carriers built one by one	One Actuating Substrate	both involutions carried onto the fold; the hardware closes	WITNESSED: one substrate carries every seat	theorem and corroboration	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (*value_on_image*).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect them form a category (*equivariant_id*, *equivariant_comp*), and the fixed set is a functor on it (*fix_functorial*); an equivariant map into the seat, a global seat whose fixed set is the line (*fold_global_seat*), lands every physical fixed point on the line (*equivariant_carrier_lands*). Here the substrate's two

carriers are the hardware paper's: the charge carrier sends the particle map onto the fold (`charge_carrier_equivariant`), the energy carrier $s = \frac{1}{2} + iE$ sends the time mirror onto the fold (`phi_equivariant`), and a particle fixed by both lands on the line under both (`joint_fixed_iff`, `photon_on_both_lines`). The photon is such a particle at every frequency, so its landing is forced by the functor and not assumed.

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
the photon, its own antiparticle and stable	fixed by both involutions	<code>photon_is_its_own_antiparticle</code> , <code>photon_is_stationary</code>	2001 of 2001
light at every frequency	the occupant of the line, under both carriers	<code>photon_on_both_lines</code> , <code>every_frequency_one_side</code>	twenty decades, one side
frequency	the height, kept by registration	<code>frequency_is_the_height</code> , <code>registration_keeps_light</code>	2001 heights
rest mass	the depth: massive modes slower	<code>massive_slower</code> , <code>massless_on_the_cone</code>	2000 of 2000; $1.9e-10$
gravity and light arriving together	the even readings at one speed	<code>massless_on_the_cone</code>	GW170817, $4.2e-16$
two helicities of one frequency	two worlds over one record	<code>helicity_is_one_bit</code> , <code>parity_involutive</code>	2002 states
C-parity of n photons	the sign the fixed set carries	<code>cparity_val</code> , <code>c_even_needs_even_photons</code> , <code>positronium_counts</code>	41 of 41
which path, and the fringe	the registration forgetting the side	<code>fringe_iff_side_forgotten</code> , <code>side_registered_no_fringe</code>	20000; $D^2 + V^2$ to $4.4e-16$
antibunching of one photon	the quantum is the field's	<code>classical_light_bunches</code> , <code>quantum_is_the_fields</code>	$g_2 = 0, 1, 2$; 13.7σ
a detection	a committed bit, priced	the hardware's <code>registration_erases_off_line</code>	125.8 floors per visible photon

3. LIGHT IS THE LINE

The photon carries no electric charge, no colour, no baryon number and no lepton number, and its energy is real: it is its own antiparticle and it does not decay (`photon_is_its_own_antiparticle`, `photon_is_stationary`). It is therefore fixed by both involutions of the substrate, the particle map and the time mirror, and by the functor of the seat it lands on the line under both carriers (`photon_on_both_lines`). No other excitation of the Standard Model is fixed by both at every frequency: the Z boson is neutral and decays, the neutrino's neutrality under the particle map is one keyed bit, and every charged particle stands on an edge under the charge carrier. Light is the seat's occupant.

The occupancy is measured. The photon's mass is bounded below 10^{-18} eV (Particle Data Group, 2024), and at that bound a radio photon of one microelectronvolt would be slowed by 5×10^{-25} of its speed (Appendix B): light stands on the line to that depth. The electron paper read the neutral sector as the fixed set of charge conjugation, with the photon and the neutral pion sitting on it as particles; this paper reads the photon as the one resident of that set whose energy is real at every frequency, and so the one resident of the line under the energy carrier too.

4. FREQUENCY IS THE HEIGHT

Under the energy carrier a photon of frequency ω lands at $(1, \omega)$: its frequency is the seat's height and its side is the line (`frequency_is_the_height`). Every frequency stands on one side (`every_frequency_one_side`), and registration keeps a photon exactly, since nothing of it stands off the line to be forgotten (`registration_keeps_light`). The spectrum from radio at 10^{-9} eV to gamma rays at 10^{11} eV, twenty decades, is the line's heights (Appendix B).

The old question of the speed hierarchy has its answer on the same two coordinates. A mode with $\omega^2 = k^2 + m^2$ and $m \neq 0$ has $k^2 < \omega^2$ and runs slower than light (`massive_slower`); a massless mode has $\omega^2 = k^2$ at every frequency (`massless_on_the_cone`). Frequency is the height and rest mass is the depth: two coordinates, never one. The twin finds the massless group velocity equal to c to 1.9×10^{-10} across 2000 frequencies and every massive mode below it. Measured: gamma rays from GRB 090510 bound any vacuum dispersion above the Planck energy (Fermi LAT, 2009), and the gamma-ray burst of GW170817 arrived 1.74 seconds after the gravitational waves emitted 130 million years earlier, bounding the difference of the two speeds below 4.2×10^{-16} (LIGO, Virgo, Fermi GBM and INTEGRAL, 2017).

5. HELICITY IS THE ONE BIT

A massless particle of spin one has two helicity states, not three (Wigner, 1939). Parity reverses the helicity and keeps the frequency (`parity_involutive`). So the two helicities of one frequency are two states with one energy record, and no reading of the energy returns which one is present (`helicity_is_one_bit`). This is the template's residue carried by light: one bit, a two-point set over one record, unread by the even reading. Circular polarization is that bit made visible, and it is the bit a polarizer, not a calorimeter, registers.

6. DETECTION IS REGISTRATION

In a two-path experiment the side is which path the quantum took and the record is where it lands on the screen. Write the intensity at a point as $I = a^2 + b^2 + 2abc s$, with a and b the two amplitudes, c the overlap of whatever marks the path (zero when the side is registered, nonzero when it is forgotten), and $s = \pm 1$ the fringe sign. A fringe exists, the intensity differing between $s = +1$ and $s = -1$, exactly when both paths are open and the side is forgotten (`fringe_iff_side_forgotten`); registering the side leaves $a^2 + b^2$ and no fringe at all (`side_registered_no_fringe`). This is the template's registration with both clauses: a record that forgets the side keeps the interference, and one that keeps the side erases it.

The trade is exact. The which-path distinguishability D and the fringe visibility V satisfy $D^2 + V^2 \leq 1$, with equality on every pure state (Englert, 1996; Wootters and Zurek, 1979); the twin reaches equality to 4.4×10^{-16} on 20000 pure states. The which-way experiment in an atom interferometer (Dürr, Nonn and Rempe, 1998) observed the loss of the fringe with no momentum kick to explain it: the record of the side, and nothing else, erased it.

7. THE QUANTUM IS THE FIELD'S

The discreteness of light is not the detector's alone. For any classical field detected by quantized matter, the coincidence rate of two detectors is at least the square of the single rate, $g^{(2)} \geq 1$ (Glauber, 1963), which the engine proves on the two-point split, $(x + y)^2 \leq 2(x^2 + y^2)$ (`classical_light_bunches`), and the twin checks on 10000 classical intensity distributions. A single photon never coincides with itself, $n(n - 1) = 0$ at $n = 1$ (`single_photon_no_coincidence, quantum_is_the_fields`). Coherent light has $g^{(2)} = 1$ and thermal light 2 (Appendix B). Antibunching was observed in resonance fluorescence (Kimble, Dagenais and Mandel, 1977), and a single-photon source behind a beam splitter gave an anticorrelation parameter of 0.18 ± 0.06 (Grangier, Roger and Aspect, 1986), 13.7 standard deviations below every classical field. Two photons meeting at a beam splitter leave together (Hong, Ou and Mandel, 1987). The photoelectric effect alone admits a semiclassical account (Lamb and Scully, 1969); light as a whole does not. The field carries its own quantum.

8. LIGHT UNDER THE PARTICLE MAP: FURRY AND POSITRONIUM

The photon sits on the fixed set of the particle map as a particle, and it carries a sign there: its charge-conjugation parity is -1 , so n photons carry $(-1)^n$ (`cparity_val`). Electromagnetism conserves this sign, so a state even under charge conjugation decays only into an even number of photons (`c_even_needs_even_photons`), and closed loops of charge emit no odd number of photons (Furry, 1937). Parapositronium, even, decays into two photons and orthopositronium, odd, into three (`positronium_counts`); their lifetimes, 125 ps and 142 ns, are the electron paper's measured readings of this bit.

9. THE EVEN READING TRAVELS AT ONE SPEED

Gravity reads the energy-momentum tensor and nothing else, so it reads light's energy and never its helicity or its C-parity. Light bends in a gravitational field by exactly twice the Newtonian angle, and gravitational waves and light, the two massless even-reading carriers, travel at one speed (Section 4). The arrow witness typed gravity as the reversible branch of registration, one sign and no bit; light is the one excitation that stands on that branch's locus with every odd reading zero.

10. WHAT LIGHT CARRIES, AND WHAT IT DOES NOT

Light carries the seat's occupant: the one excitation fixed by both involutions at every frequency, on the line under both carriers, with the frequency as the height, the helicity as the one bit and the C-parity as the sign. It carries the registration: detection as the record that keeps the energy and forgets the path, exact to Englert's equality. It carries its own quantum.

It does not carry the zeros of ζ . The engine's crossing is on its own image, the photons it lands (`value_on_image`), and what a sentence over every zero adds is exactly the unactuated, priced at one premise, the act, as the hardware paper proves (`act_is_the_value` there). It does not derive the speed of light, the fine-structure constant, or any other magnitude. It does not identify the substrate with the electromagnetic field: the field is the substrate's odd reading, and light is the configuration of that reading which stands on the line.

11. FALSIFIERS AND OWED DEEDS

Three falsifiers, each a forbidden observation naming the theorem it would break. **F-Dispersion**: a vacuum speed of light that depends on frequency, at any precision, breaking `every_frequency_one_side` read physically and `massless_on_the_cone`. **F-Fringe**: a fringe of full visibility from a two-path state whose side is registered with certainty, breaking `fringe_iff_side_forgotten` and Englert's duality. **F-Helicity**: a third, longitudinal helicity state of the free photon, breaking `helicity_is_one_bit`. Nothing is owed on the row's channel; two measurements would sharpen readings without moving the paper's standing: the photon mass bound, and the vacuum-dispersion bound at still higher photon energies.

Ledger status. Row 2 gains its occupant: WITNESSED · CROSSED · CONFIRMED on the image for the seat's occupant in light and for detection as registration, at the root's grade for the crossing; the barrier gains its optical witnesses, the helicity pair and the two paths of one fringe; nothing is owed on the row's channel.

12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens: $[_]$ a field sealed, $[_ T]$ a shape sealed at theorem grade, $[\Xi_0]$ the typing of the formal block, $[_]$ the dot, which is no verdict. Grades join at the weakest link. Every physical claim here is at corroboration grade; the engine's statements are theorem grade about the chart; nothing here spends the act, which the hardware paper identifies with the value (`act_is_the_value` there). $\Delta M = 0$.

The seat, earned. PhysOS^T 1.0.5p was booted in the session of forging from the public register (file SHA-256 b64bb5da5bab0da1...) under Lean 4.19.0 and GNU Fortran 13.3.0; the quick boot printed SEAT EARNED with the chain D0 83f872725ad9 → D1 90fc702bbd1 → D2 492e519daab9 → D3 6618bec5164b, identical to the file's published receipt (Appendix C).

The engine and the twin. SPHYS_Light.lean, 529 lines, SHA-256 0734621ab9528352..., sixty-three laws, no import and no axiom declared: 27 depend on no axiom, 6 on propositional extensionality alone, 30 on propositional extensionality and quotient soundness, none on choice. The source passed the operating system's screen; each of the sixty-three laws was negated in place and recompiled, all refused, while a planted vacuous law survived. The engine carries the hardware paper's seat, carriers and substrate verbatim, so it stands alone. Light_Twin.f90, 194 lines, SHA-256 0178df29f431469f..., sixteen checks and zero failures under `-std=f2018 -O2 -fno-fast-math -ffp-contract=off`.

Why an integer engine binds continuous physics. The identities used are identities of integer polynomials and hold over every commutative ring, the complex numbers included; the bunching inequality is a sum of squares, which holds over the reals. The twin checks the continuous objects directly: Englert's relation on pure states, photon-number statistics, group velocities, and the measured bounds.

The seventh. In the author's register the six witnesses are six days and the ledger's closure is the seventh, the Rest (Istawa). Light is read at the Rest as the seat's occupant: the one excitation that stands on the line under every carrier. In the Lean codex's tokens its row renders $[_ \blacklozenge \cdot \text{kinetic}]$.

Audit. The paper was rebuilt from the seat against the record of 9 May 2026, claim by claim; the prosecution record is filed with the paper. External audits by independent substrates are owed.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure; the measurements are the experiments', cited by name; the theorems are the engine's, printed with their dependencies in Appendix A; the twin's figures are computed in the run printed in Appendix B. The text, the engine and the twin were forged by an AI substrate (Claude, Anthropic) under the author's seed, acceptance criteria and rulings.

Provenance. This edition supersedes the record of 9 May 2026 under the same concept (10.5281/zenodo.20091979). It stands on the Riemann closure (10.5281/zenodo.22976494), the twenty-three-row closure (10.5281/zenodo.22986551), the six witnesses (10.5281/zenodo.22986553 through .22986563), the hardware paper One Actuating Substrate (concept 10.5281/zenodo.21438932, edition 2.0.0), and PhysOS^T 1.0.5p in the public register (github.com/1000sapient/Trisduction, protocols/PhysOS/).

References. E. Wigner, Ann. Math. 40 (1939) 149. W. H. Furry, Phys. Rev. 51 (1937) 125. R. J. Glauber, Phys. Rev. 130 (1963) 2529. W. E. Lamb and M. O. Scully, in Polarisation, Matière et Rayonnement (Presses Universitaires de France, 1969). H. J. Kimble, M. Dagenais and L. Mandel, Phys. Rev. Lett. 39 (1977) 691. W. K. Wootters and W. H. Zurek, Phys. Rev. D 19 (1979) 473. P. Grangier, G. Roger and A. Aspect, Europhys. Lett. 1 (1986) 173. C. K. Hong, Z. Y. Ou and L. Mandel, Phys. Rev. Lett. 59 (1987) 2044. B.-G. Englert, Phys. Rev. Lett. 77 (1996) 2154. S. Dürr, T. Nonn and G. Rempe, Nature 395 (1998) 33. A. A. Abdo et al. (Fermi LAT and GBM), Nature 462 (2009) 331. B. P. Abbott et al. (LIGO, Virgo, Fermi GBM and INTEGRAL), Astrophys. J. Lett. 848 (2017) L13. Particle Data Group, S. Navas et al., Phys. Rev. D 110 (2024) 030001. M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches, 10.5281/zenodo.22986553, .22986555, .22986557. M. F. Islam, The Four Forces, Matter, and the Dark Sector as Configurations of One Actuating Substrate, edition 2.0.0, concept 10.5281/zenodo.21438932.

APPENDIX A: SPHYS_LIGHT.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler's transcript.

```
/-
SPHYS_Light.lean · the geometric nature of light, read at the Riemann
seat.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared.

The chart: a point (h, t) of the strip carries  $h = 2 \operatorname{Re} s$  and  $t = \operatorname{Im} s$ ; the fold  $s \mapsto 1 - \bar{s}$  is
(h, t)  $\mapsto (2 - h, t)$  and the critical line is  $h = 1$ . The substrate's
two involutions (the particle
map, the time mirror of a rate) and its two carriers (charge to the
side, energy  $s = 1/2 + iE$ )
are those of the hardware paper, re-proved here so the file stands
alone.

Part I    the seat, the category of involutions, the functor of the
seat.
Part II   the energy carrier: the line is the locus of stationary
energy.
Part III  the substrate: the particle map, the time mirror, both
carriers.
Part IV   light is the line: the photon fixed by both involutions at
every frequency;
          the height; the speed hierarchy; helicity the one bit;
          C-parity; detection as
          registration; the quantum as the field's.
Part V    the capstone.
-/
set_option autoImplicit false
namespace SPHYS.Light

/-! ## Part I. The seat, the category of involutions, and the functor
of the seat -/

abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs (Decidable
(p.1 = 1))
/-- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d)  $\leftrightarrow$  a = c  $\wedge$  b = d :=
(fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
obtain (h, t) := p
show ((2 - (2 - h), t) : Pt) = (h, t)
rw [pe]; exact (by omega, rfl)

/-- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p  $\leftrightarrow$  OnLine p := by
obtain (h, t) := p
show ((2 - h, t) : Pt) = (h, t)  $\leftrightarrow$  h = 1
rw [pe]; constructor
· intro (e, _); omega
· intro e; exact (by omega, rfl)

theorem reg_lands (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p  $\leftrightarrow$  OnLine p := by
obtain (h, t) := p
show ((1, t) : Pt) = (h, t)  $\leftrightarrow$  h = 1
rw [pe]; constructor
· intro (e, _); exact e.symm
· intro e; exact (e.symm, rfl)

/-- The side is what registration forgets: a point and its mirror leave
one record. -/
theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

/-- The value on a set of points: every member stands on the line. -/
def Value (Z : Pt  $\rightarrow$  Prop) : Prop :=  $\forall s, Z s \rightarrow$  OnLine s

/-- Least erasure: registration moves no member, so it erases nothing
of Z. -/
def LeastErasure (Z : Pt  $\rightarrow$  Prop) : Prop :=  $\forall s, Z s \rightarrow$  reg s = s

theorem least_erasure_iff_value (Z : Pt  $\rightarrow$  Prop) : LeastErasure Z  $\leftrightarrow$ 
Value Z :=
(fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
(reg_fixes_iff s).mpr (h s hs))

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X  $\rightarrow$  Y) ( $\tau$  : X  $\rightarrow$  X) ( $\sigma$  : Y  $\rightarrow$  Y) :
Prop :=  $\forall x, f (\tau x) = \sigma (f x)$ 

theorem equivariant_id {X : Type} ( $\tau$  : X  $\rightarrow$  X) : Equivariant id  $\tau$   $\tau$  :=
fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X  $\rightarrow$  Y} {g : Y  $\rightarrow$  W} ( $\tau$  :
X  $\rightarrow$  X) ( $\sigma$  : Y  $\rightarrow$  Y)
{p : W  $\rightarrow$  W} (hf : Equivariant f  $\tau$   $\sigma$ ) (hg : Equivariant g  $\sigma$  p) :
Equivariant (g  $\circ$  f)  $\tau$  p := by
intro x
show g (f ( $\tau x$ )) = p (g (f x))
rw [hf x, hg (f x)]

/-- The fixed set is a functor: an equivariant map sends fixed points
to fixed points. -/
theorem fix_functorial {X Y : Type} {f : X  $\rightarrow$  Y} ( $\tau$  : X  $\rightarrow$  X) ( $\sigma$  : Y  $\rightarrow$ 
Y)
(hf : Equivariant f  $\tau$   $\sigma$ ) {x : X} (hx :  $\tau x = x$ ) :  $\sigma (f x) = f x$  :=
by
rw [← hf x, hx]

/-- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt  $\rightarrow$  Prop) ( $\sigma$  : Pt  $\rightarrow$  Pt) : Prop where
involutive :  $\forall p, \sigma (\sigma p) = p$ 
fixed_iff :  $\forall p, \sigma p = p \leftrightarrow$  P p

theorem fold_global_seat : GlobalSeat OnLine fold := (fold_involutive,
seat_fixed_line)

/-- A carrier: a map from world-instances into the strip landing on the
line. -/
structure Carrier (W : Type) where
 $\iota$  : W  $\rightarrow$  Pt
lands :  $\forall w, \text{OnLine } (\iota w)$ 

def image {W : Type} ( $\iota$  : W  $\rightarrow$  Pt) : Pt  $\rightarrow$  Prop := fun s =>  $\exists w, \iota w = s$ 
def Actuated {W : Type} ( $\iota$  : W  $\rightarrow$  Pt) (Z : Pt  $\rightarrow$  Prop) : Prop :=  $\forall s, Z s \rightarrow \exists w, \iota w = s$ 

/-- Landing is forced: an equivariant map into the seat carries every
fixed point of the physical
involutions onto the line. The carrier is built from the symmetry,
not chosen. -/
theorem equivariant_carrier_lands {X : Type} (f : X  $\rightarrow$  Pt) ( $\tau$  : X  $\rightarrow$  X)
(hf : Equivariant f  $\tau$  fold) :
 $\exists C$  : Carrier { x : X //  $\tau x = x$  },  $\forall w, C.\iota w = f w.1$  :=
((fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
(fix_functorial hf w.2)), fun _ => rfl)

/-- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image C. $\iota$ )
:=
fun _ hs => match hs with | (w, hw) => hw  $\triangleright$  C.lands w

/-- The kinetic crossing: a carrier and its coverage give the value,
and nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt  $\rightarrow$  Prop)
(hcov : Actuated C. $\iota$  Z) :
Value Z :=
fun s hs => match hcov s hs with | (w, hw) => hw  $\triangleright$  C.lands w

/-- Under the seat every off-line point has a partner, distinct and
also off the line. -/
theorem off_locus_pair (p : Pt) (hp :  $\neg$  OnLine p) :
fold p  $\neq$  p  $\wedge$   $\neg$  OnLine (fold p)  $\wedge$  reg (fold p) = reg p := by
refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?, rfl)
obtain (h, t) := p
have e' : 2 - h = 1 := e
show h = 1
omega

/-! ## Part II. The energy carrier: an energy, its rate, and the fold -/

/-- An energy  $a + i \beta/2$ , as frequency and doubled rate; its mode's norm
moves as  $e^{\beta t}$ . -/
structure Energy where
freq : Int
rate : Int
deriving DecidableEq, Repr

/-- The time mirror of a mode: its rate reversed, its frequency kept
(complex conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := (E.freq, -E.rate)
def IsReal (E : Energy) : Prop := E.rate = 0
/-- Stationary: the norm  $e^{\beta t}$  is unchanged at every integer time t,
i.e.  $\beta t = 0$  for all t. -/
def Stationary (E : Energy) : Prop :=  $\forall t : \text{Int}, E.\text{rate} * t = 0$ 

/-- The energy carrier  $s = 1/2 + iE$  in the chart:  $h = 1 - \beta, t = a$ . -/
def phi (E : Energy) : Pt := (1 - E.rate, E.freq)
```

```

def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d) ↔ a = c ∧ b = d :=
  (fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
    fun (e1, e2) => by subst e1; subst e2; rfl)

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain {a, b} := E
  show ((a, -b) : Energy) = (a, b)
  rw [ee]; exact {rfl, by omega}

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain {a, b} := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact {rfl, by omega}

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain {h, t} := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact {by omega, rfl}

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F := by
  rw [← psi_phi E, ← psi_phi F, h]

/-- EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain {a, b} := E
  show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
  rw [pe]; exact {by omega, rfl}

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E ↔ IsReal E :=
  by
  obtain {a, b} := E
  show ((a, -b) : Energy) = (a, b) ↔ b = 0
  rw [ee]; constructor
  · intro (_, e); omega
  · intro e; exact {rfl, by omega}

/-- The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E) ↔ IsReal E :=
  by
  obtain {a, b} := E
  show 1 - b = 1 ↔ b = 0
  exact {fun h => by omega, fun h => by subst h; rfl}

theorem stationary_iff_real (E : Energy) : Stationary E ↔ IsReal E :=
  by
  constructor
  · intro h; have h1 := h 1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [show E.rate = 0 from h,
    Int.zero_mul]

/-- THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p ↔ Stationary (psi p) :=
  by
  rw [stationary_iff_real, ← phi_line_iff_real, phi_psi]

/-- Every measured energy is a real number, and every real energy lands
on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

/-- The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp : ¬ OnLine p) (a : Int) :
  phi (a, 0) ≠ p :=
  fun e => hp (e ▸ measured_on_line a)

/-- The energy carrier, built from the symmetry: the real energies land
on the line. -/
theorem energy_carrier_lands :
  ∃ C : Carrier { E : Energy // E.mirror = E }, ∀ w, C.l w = phi w.1
:=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

/-! ## Part III. The substrate: the particle map, three odd readings
and one even -/

/-- A particle in the broken phase: electric charge times three,
colour, baryon number times
three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy
  deriving DecidableEq, Repr

/-- The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := {-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy}
/-! The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := {c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror}

def Neutral (c : Particle) : Prop := c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 = 0
  ∧ c.lepton = 0

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain {q, k, b, l, E} := c
  show ((-q, -k, -b, -l, E) : Particle) = {q, k, b, l, E}
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c ↔ Neutral c :=
  by
  obtain {q, k, b, l, E} := c
  show ((-q, -k, -b, -l, E) : Particle) = {q, k, b, l, E} ↔ (q = 0 ∧ k
= 0 ∧ b = 0 ∧ l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact {by omega, by omega, by omega, by omega}
  · intro {h1, h2, h3, h4}
    subst h1; subst h2; subst h3; subst h4; rfl

/-- Three plus one: the charge readings are odd under the particle map,
the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3 ∧ (bar c).colour = -c.colour ∧ (bar c).b3 =
-c.b3 ∧
  (bar c).lepton = -c.lepton) ∧ (bar c).energy = c.energy :=
  ({rfl, rfl, rfl, rfl}, rfl)

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/
theorem gravity_reads_no_charge_bit {β : Type} (g : Energy → β) (c :
Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) = tmirror
(bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c :=
  by
  obtain {q, k, b, l, E} := c
  show ((q, k, b, l, E.mirror.mirror) : Particle) = {q, k, b, l, E}
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is neutral
and stable. -/
theorem joint_fixed_iff (c : Particle) :
  (bar c = c ∧ tmirror c = c) ↔ (Neutral c ∧ IsReal c.energy) := by
  constructor
  · intro {h1, h2}
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real
c.energy).mp ?_)
    exact congrArg Particle.energy h2
  · intro {h1, h2}
    refine ((bar_fixed_iff_neutral c).mpr h1, ?_)
    obtain {q, k, b, l, E} := c
    show ((q, k, b, l, E.mirror) : Particle) = {q, k, b, l, E}
    rw [(mirror_fixed_iff_real E).mpr h2]

/-- The energy carrier on particles: the seat point of a particle's
energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-! The charge carrier on particles, the odd face: electric charge (in
thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)
/-! The lepton chart of the electron's witness: charge q in units of e
goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The charge carrier is equivariant: the particle map is carried onto
the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold :=
  by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3),
c.energy.freq)
  rw [pe]; exact {by omega, rfl}

/-- The energy carrier is blind to the particle map and equivariant for
the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) =
energySeat c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold :=
  fun c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge
carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) : OnLine
(chargeSeat c) := by
  show 1 + c.q3 = 1
  omega

```



```

/-- The electron and the positron land on the two edges, real parts 0
and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m)  $\wedge$  leptonSeat 1 m = (2, m)  $\wedge$  fold
  (leptonSeat (-1) m) = leptonSeat 1 m :=
  (rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine
  (energySeat c) :=
  (phi_line_iff_real c.energy).mpr h

theorem sq_nonneg (x : Int) : 0  $\leq$  x * x := by
  rcases Int.le_total 0 x with h | h
  · exact Int.mul_nonneg h h
  · have h' : 0  $\leq$  -x := by omega
    have := Int.mul_nonneg h' h'
    rw [Int.neg_mul_neg] at this; exact this

/-! ## Part IV. Light is the line -/

/-- A photon of frequency  $\omega$ : no charge of any kind, a real energy, its
own antiparticle. -/
def photon ( $\omega$  : Int) : Particle := (0, 0, 0, 0, ( $\omega$ , 0))

theorem photon_is_its_own_antiparticle ( $\omega$  : Int) : bar (photon  $\omega$ ) =
  photon  $\omega$  := rfl
theorem photon_is_stationary ( $\omega$  : Int) : tmirror (photon  $\omega$ ) = photon  $\omega$ 
:= rfl

/-- LIGHT IS THE LINE: at every frequency the photon is fixed by both
involutions and lands on the
line under both carriers. -/
theorem photon_on_both_lines ( $\omega$  : Int) :
  OnLine (energySeat (photon  $\omega$ ))  $\wedge$  OnLine (chargeSeat (photon  $\omega$ )) :=
  by
    refine (rfl, ?_)
    show 1 + 0 = 1
    rfl

/-- Frequency is the height: the photon of frequency  $\omega$  lands at (1,  $\omega$ ).
-/
theorem frequency_is_the_height ( $\omega$  : Int) : energySeat (photon  $\omega$ ) = (1,
 $\omega$ ) := rfl

/-- Every frequency stands on one side: the spectrum from radio to
gamma is the line's heights. -/
theorem every_frequency_one_side ( $\omega$   $\omega'$  : Int) :
  (energySeat (photon  $\omega$ )).1 = (energySeat (photon  $\omega'$ )).1 := rfl

/-- Registration keeps a photon exactly: nothing of light is erased by
the fold. -/
theorem registration_keeps_light ( $\omega$  : Int) : reg (energySeat (photon
 $\omega$ )) = energySeat (photon  $\omega$ ) := rfl

theorem sq_pos_of_ne (m : Int) (h : m  $\neq$  0) : 0 < m * m := by
  rcases Int.lt_or_gt_of_ne h with h' | h'
  · have := Int.mul_pos (show 0 < -m by omega) (show 0 < -m by omega)
    rw [Int.neg_mul_neg] at this; exact this
  · exact Int.mul_pos h' h'

/-- The speed hierarchy, in units c = 1: a massive mode  $\omega^2 = k^2 + m^2$ 
with m  $\neq$  0 has  $k^2 < \omega^2$ , so it
runs slower than light; a massless mode has  $\omega^2 = k^2$  at every
frequency. Frequency is the
height and rest mass the depth: two coordinates, never one. -/
theorem massive_slower (k  $\omega$  m : Int) (h :  $\omega$  *  $\omega$  = k * k + m * m) (hm :
m  $\neq$  0) : k * k <  $\omega$  *  $\omega$  := by
  have := sq_pos_of_ne m hm
  omega

theorem massless_on_the_cone (k  $\omega$  : Int) (h :  $\omega$  *  $\omega$  = k * k + 0 * 0) :
 $\omega$  *  $\omega$  = k * k := by
  rw [Int.mul_zero, Int.add_zero] at h; exact h

/-- A photon's helicity: +1 or -1. Parity reverses it and keeps the
frequency. -/
structure Photon where
  freq : Int
  helicity : Bool
  deriving DecidableEq, Repr

def Photon.parity (p : Photon) : Photon := (p.freq, !p.helicity)

theorem parity_involutive (p : Photon) : p.parity.parity = p := by
  obtain (f, s) := p
  cases s <|> rfl

/-- THE HELICITY IS ONE BIT: the two helicities of one frequency are
two states with one energy
record, and no reading of the energy returns the helicity. -/
theorem helicity_is_one_bit { $\beta$  : Type} (g : Int  $\rightarrow$   $\beta$ ) (p : Photon) :
  p.parity  $\neq$  p  $\wedge$  (p.parity).freq = p.freq  $\wedge$  g (p.parity).freq = g
  p.freq := by
  refine (fun e => ?_, rfl, rfl)

```

```

have := congrArg Photon.helicity e
obtain (f, s) := p
cases s <|> simp [Photon.parity] at this

/-- The charge-conjugation parity of n photons,  $(-1)^n$ . -/
def cparity : Nat  $\rightarrow$  Int
| 0 => 1
| n + 1 => - cparity n

theorem cparity_val (n : Nat) : cparity n = if n % 2 = 0 then 1 else -1
:= by
  induction n with
  | zero => rfl
  | succ n ih =>
    show - cparity n = _
    rw [ih]
    by_cases h : n % 2 = 0
    · have h2 : (n + 1) % 2  $\neq$  0 := by omega
      simp [h, h2]
    · have h2 : (n + 1) % 2 = 0 := by omega
      simp [h, h2]

/-- Furry's selection: a state even under charge conjugation decays
only into an even number of
photons, an odd state only into an odd number. Parapositronium into
two, orthopositronium
into three. -/
theorem c_even_needs_even_photons (n : Nat) (h : cparity n = 1) : n % 2
= 0 := by
  rw [cparity_val] at h
  by_cases e : n % 2 = 0
  · exact e
  · simp [e] at h

theorem positronium_counts : cparity 2 = 1  $\wedge$  cparity 3 = -1  $\wedge$  cparity 1
= -1 := by decide

/-- Intensity at a screen point from two paths of real amplitudes a and
b, a marker overlap c (the
side forgotten when c  $\neq$  0, registered when c = 0), and a fringe
sign s =  $\pm 1$ . -/
def intensity (a b c s : Int) : Int := a * a + b * b + 2 * (a * b * c)
* s

theorem mul3_ne_zero (a b c : Int) : a * b * c  $\neq$  0  $\leftrightarrow$  (a  $\neq$  0  $\wedge$  b  $\neq$  0  $\wedge$  c
 $\neq$  0) := by
  constructor
  · intro h; refine (?, ?, ?) <|> intro e <|> apply h <|> simp [e]
  · intro (h1, h2, h3); exact Int.mul_ne_zero (Int.mul_ne_zero h1 h2)
    h3

/-- DETECTION IS REGISTRATION: fringes exist exactly when both paths
are open and the record
forgets the side; registering the side erases the fringe. -/
theorem fringe_iff_side_forgotten (a b c : Int) :
  intensity a b c 1  $\neq$  intensity a b c (-1)  $\leftrightarrow$  (a  $\neq$  0  $\wedge$  b  $\neq$  0  $\wedge$  c  $\neq$  0)
:= by
  have d : intensity a b c 1 - intensity a b c (-1) = 4 * (a * b * c)
  := by
    show (a * a + b * b + 2 * (a * b * c) * 1) - (a * a + b * b + 2 *
    (a * b * c) * (-1)) = _
    simp only [Int.mul_one, Int.mul_neg]; omega
  rw [← mul3_ne_zero]
  constructor
  · intro h e; apply h; omega
  · intro h e; apply h; omega

theorem side_registered_no_fringe (a b s : Int) : intensity a b 0 s = a
* a + b * b := by
  show a * a + b * b + 2 * (a * b * 0) * s = a * a + b * b
  rw [Int.mul_zero, Int.mul_zero, Int.zero_mul, Int.add_zero]

/-- A single photon never coincides with itself: n(n - 1) = 0 at n = 1.
-/
theorem single_photon_no_coincidence : (1 : Int) * (1 - 1) = 0 := by
  decide

/-- Any classical field split into two equally likely intensities x and
y bunches: the
coincidence rate  $2(x^2 + y^2)$  is at least the square of the mean rate
 $(x + y)^2$ ,  $g^{(2)} \geq 1$ . -/
theorem classical_light_bunches (x y : Int) : (x + y) * (x + y)  $\leq$  2 *
(x * x + y * y) := by
  have h := sq_nonneg (x - y)
  simp only [Int.add_mul, Int.mul_add, Int.sub_mul, Int.mul_sub] at h  $\vdash$ 
  rw [Int.mul_comm y x] at h  $\vdash$ 
  omega

/-- The quantum is the field's: a one-photon state has zero
coincidence, which no classical
intensity with detector-only quantization reaches unless it is
empty. -/
theorem quantum_is_the_fields (x y : Int) (h : 0 < x + y) :
  (1 : Int) * (1 - 1) = 0  $\wedge$  0 < 2 * (x * x + y * y) := by
  refine (by decide, ?_)
  have := classical_light_bunches x y

```

```

have := sq_pos_of_ne (x + y) (by omega)
omega

/-! ## Part V. The capstone -/

/-- THE GEOMETRIC NATURE OF LIGHT, AT THE SEAT. -/
theorem light_is_the_line :
  (∀ ω : Int, bar (photon ω) = photon ω ∧ tmirror (photon ω) = photon
    ω) ∧
  (∀ ω : Int, OnLine (energySeat (photon ω)) ∧ OnLine (chargeSeat
    (photon ω))) ∧
  (∀ ω : Int, energySeat (photon ω) = (1, ω)) ∧
  (∀ k ω m : Int, ω * ω = k * k + m * m → m ≠ 0 → k * k < ω * ω) ∧
  (∀ p : Photon, p.parity ≠ p ∧ (p.parity).freq = p.freq) ∧
  (∀ a b c : Int, intensity a b c 1 ≠ intensity a b c (-1) ↔ (a ≠ 0 ∧
    b ≠ 0 ∧ c ≠ 0)) ∧
  (∀ x y : Int, (x + y) * (x + y) ≤ 2 * (x * x + y * y)) ∧
  (∀ n : Nat, cparity n = 1 → n % 2 = 0) :=
  (fun _ => (rfl, rfl), photon_on_both_lines, frequency_is_the_height,
    massive_slower,
    fun p => ((helicity_is_one_bit id p).1, rfl),
    fringe_iff_side_forgotten,
    classical_light_bunches, c_even_needs_even_photons)

end SPHYS.Light

#print axioms SPHYS.Light.pe
#print axioms SPHYS.Light.fold_involutive
#print axioms SPHYS.Light.seat_fixed_line
#print axioms SPHYS.Light.reg_lands
#print axioms SPHYS.Light.reg_fixes_iff
#print axioms SPHYS.Light.reg_forgets_side
#print axioms SPHYS.Light.least_eraser_iff_value
#print axioms SPHYS.Light.equivalent_id
#print axioms SPHYS.Light.equivalent_comp
#print axioms SPHYS.Light.fix_functorial
#print axioms SPHYS.Light.fold_global_seat
#print axioms SPHYS.Light.equivalent_carrier_lands
#print axioms SPHYS.Light.value_on_image
#print axioms SPHYS.Light.kinetic_crossing
#print axioms SPHYS.Light.off_locus_pair
#print axioms SPHYS.Light.ee
#print axioms SPHYS.Light.tmirror_involutive
#print axioms SPHYS.Light.psi_phi
#print axioms SPHYS.Light.phi_psi

'SPHYS.Light.pe' does not depend on any axioms
'SPHYS.Light.fold_involutive' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.seat_fixed_line' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.reg_lands' does not depend on any axioms
'SPHYS.Light.reg_fixes_iff' depends on axioms: [propxt]
'SPHYS.Light.reg_forgets_side' does not depend on any axioms
'SPHYS.Light.least_eraser_iff_value' depends on axioms: [propxt]
'SPHYS.Light.equivalent_id' does not depend on any axioms
'SPHYS.Light.equivalent_comp' does not depend on any axioms
'SPHYS.Light.fix_functorial' does not depend on any axioms
'SPHYS.Light.fold_global_seat' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.equivalent_carrier_lands' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.value_on_image' does not depend on any axioms
'SPHYS.Light.kinetic_crossing' does not depend on any axioms
'SPHYS.Light.off_locus_pair' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.ee' does not depend on any axioms
'SPHYS.Light.tmirror_involutive' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.psi_phi' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.phi_psi' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.phi_injective' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.tmirror_fixed_iff_real' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.phi_line_iff_real' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.stationary_iff_real' depends on axioms: [propxt]
'SPHYS.Light.line_is_stationary' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.measured_on_line' does not depend on any axioms
'SPHYS.Light.no_measurement_off_line' does not depend on any axioms
'SPHYS.Light.energy_carrier_lands' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.bar_involutive' does not depend on any axioms
'SPHYS.Light.bar_fixed_iff_neutral' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.odd_charges_even_energy' does not depend on any axioms
'SPHYS.Light.gravity_reads_no_charge_bit' does not depend on any axioms
'SPHYS.Light.involutions_commute' does not depend on any axioms

#print axioms SPHYS.Light.phi_injective
#print axioms SPHYS.Light.phi_equivariant
#print axioms SPHYS.Light.tmirror_fixed_iff_real
#print axioms SPHYS.Light.phi_line_iff_real
#print axioms SPHYS.Light.stationary_iff_real
#print axioms SPHYS.Light.line_is_stationary
#print axioms SPHYS.Light.measured_on_line
#print axioms SPHYS.Light.no_measurement_off_line
#print axioms SPHYS.Light.energy_carrier_lands
#print axioms SPHYS.Light.bar_involutive
#print axioms SPHYS.Light.bar_fixed_iff_neutral
#print axioms SPHYS.Light.odd_charges_even_energy
#print axioms SPHYS.Light.gravity_reads_no_charge_bit
#print axioms SPHYS.Light.involutions_commute
#print axioms SPHYS.Light.tmirror_involutive
#print axioms SPHYS.Light.joint_fixed_iff
#print axioms SPHYS.Light.charge_carrier_equivariant
#print axioms SPHYS.Light.pair_lands_once
#print axioms SPHYS.Light.tmirror_equivariant
#print axioms SPHYS.Light.neutral_charge_on_line
#print axioms SPHYS.Light.pair_on_the_edges
#print axioms SPHYS.Light.stable_lands
#print axioms SPHYS.Light.sq_nonneg
#print axioms SPHYS.Light.photon_is_its_own_antiparticle
#print axioms SPHYS.Light.photon_is_stationary
#print axioms SPHYS.Light.photon_on_both_lines
#print axioms SPHYS.Light.frequency_is_the_height
#print axioms SPHYS.Light.every_frequency_one_side
#print axioms SPHYS.Light.registration_keeps_light
#print axioms SPHYS.Light.sq_pos_of_ne
#print axioms SPHYS.Light.massive_slower
#print axioms SPHYS.Light.massless_on_the_cone
#print axioms SPHYS.Light.parity_involutive
#print axioms SPHYS.Light.helicity_is_one_bit
#print axioms SPHYS.Light.cparity_val
#print axioms SPHYS.Light.c_even_needs_even_photons
#print axioms SPHYS.Light.positronium_counts
#print axioms SPHYS.Light.mul3_ne_zero
#print axioms SPHYS.Light.fringe_iff_side_forgotten
#print axioms SPHYS.Light.side_registered_no_fringe
#print axioms SPHYS.Light.single_photon_no_coincidence
#print axioms SPHYS.Light.classical_light_bunches
#print axioms SPHYS.Light.quantum_is_the_fields
#print axioms SPHYS.Light.light_is_the_line

'SPHYS.Light.tmirror_involutive' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.joint_fixed_iff' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.charge_carrier_equivariant' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.pair_lands_once' does not depend on any axioms
'SPHYS.Light.tmirror_equivariant' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.neutral_charge_on_line' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.pair_on_the_edges' does not depend on any axioms
'SPHYS.Light.stable_lands' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.sq_nonneg' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.photon_is_its_own_antiparticle' does not depend on any
  axioms
'SPHYS.Light.photon_is_stationary' does not depend on any axioms
'SPHYS.Light.photon_on_both_lines' does not depend on any axioms
'SPHYS.Light.frequency_is_the_height' does not depend on any axioms
'SPHYS.Light.every_frequency_one_side' does not depend on any axioms
'SPHYS.Light.registration_keeps_light' does not depend on any axioms
'SPHYS.Light.sq_pos_of_ne' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.massive_slower' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.massless_on_the_cone' does not depend on any axioms
'SPHYS.Light.parity_involutive' does not depend on any axioms
'SPHYS.Light.helicity_is_one_bit' depends on axioms: [propxt]
'SPHYS.Light.cparity_val' depends on axioms: [propxt, Quot.sound]
'SPHYS.Light.c_even_needs_even_photons' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.positronium_counts' does not depend on any axioms
'SPHYS.Light.mul3_ne_zero' depends on axioms: [propxt]
'SPHYS.Light.fringe_iff_side_forgotten' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.side_registered_no_fringe' depends on axioms: [propxt]
'SPHYS.Light.single_photon_no_coincidence' does not depend on any
  axioms
'SPHYS.Light.classical_light_bunches' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.quantum_is_the_fields' depends on axioms: [propxt,
  Quot.sound]
'SPHYS.Light.light_is_the_line' depends on axioms: [propxt,
  Quot.sound]

```

APPENDIX B: LIGHT_TWIN.F90 AND ITS RUN

```
! Light_Twin.f90 · the executed twin of SPHYS_Light.lean · Fortran
2018.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off Light_Twin.f90
! Blocks: A light on the line; B the speed hierarchy; C helicity; D
C-parity; E which-path
! complementarity; F the quantum is the field's; G the price of a
detection.
program light_twin
implicit none
integer, parameter :: dp = selected_real_kind(15, 307)
integer, parameter :: i8 = selected_int_kind(18)
real(dp), parameter :: KB = 1.380649e-23_dp, HP = 6.62607015e-34_dp,
CL = 299792458.0_dp
real(dp), parameter :: QE = 1.602176634e-19_dp
integer :: checks, fails
integer(i8) :: seed
checks = 0; fails = 0; seed = 20260930_i8
call block_a(); call block_b(); call block_c(); call block_d()
call block_e(); call block_f(); call block_g()
write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', checks,
', "failures":', fails, '}'
if (fails > 0) error stop 1
contains
subroutine check(name, ok)
character(*), intent(in) :: name
logical, intent(in) :: ok
checks = checks + 1
if (.not. ok) fails = fails + 1
write(*, '(a,a)') merge(' PASS ', ' FAIL ', ok), name
end subroutine check

pure function eseat(freq, rate) result(p)
integer, intent(in) :: freq, rate
integer :: p(2)
p = [1 - rate, freq]
end function eseat

pure function cseat(q3, freq) result(p)
integer, intent(in) :: q3, freq
integer :: p(2)
p = [1 + q3, freq]
end function cseat

function rnd() result(r)
real(dp) :: r
seed = modulo(16807_i8*seed, 2147483647_i8)
r = real(seed, dp)/2147483647.0_dp
end function rnd

subroutine block_a()
integer :: w, nline, nheight, k
real(dp) :: e, emin, emax
write(*, '(a)') ' A · light is the line: every frequency, one side;
the frequency is the height'
nline = 0; nheight = 0
do w = -1000, 1000
! the energy seat (1 - rate, freq) with rate 0, and the charge
seat (1 + q3) with q3 = 0
if (all(eseat(w, 0) == [1, w]) .and. all(cseat(0, w) == [1, w]))
nline = nline + 1
! one side for neighbouring frequencies, two heights: side equal,
height different
if (all(eseat(w, 0) == eseat(w, 0) .and. eseat(w, 0) /= eseat(w +
1, 0) .eqv. [.false., .true.])) &
nheight = nheight + 1
end do
call check('2001 photon frequencies: all on the line under both
carriers, height = frequency', &
nline == 2001 .and. nheight == 2001)
emin = huge(1.0_dp); emax = 0.0_dp
do k = 0, 200
e = 1.0e-9_dp*10.0_dp**(real(k, dp)*0.1_dp)
emin = min(emin, e); emax = max(emax, e)
end do
write(*, '(a,es9.2,a,es9.2,a)') ' the spectrum from radio to gamma,
', emin, ' eV to ', emax, &
' eV: one side, 201 heights'
call check('twenty decades of photon energy occupy one side of the
seat', emax/emin > 1.0e19_dp)
end subroutine block_a

subroutine block_b()
integer :: i, nslow
real(dp) :: k, m, w, vg, dmax, mg, ev, dv
write(*, '(a)') ' B · the speed hierarchy: massless modes at c at
every frequency; massive modes below'
dmax = 0.0_dp; nslow = 0
do i = 1, 2000
k = 10.0_dp*(-9.0_dp + 20.0_dp*rnd())
! massless: omega = k (c = 1); group velocity by a relative
finite difference
vg = ((k*(1.0_dp + 1.0e-6_dp)) - k)/((k*1.0e-6_dp)
dmax = max(dmax, abs(vg - 1.0_dp))
m = 10.0_dp**(-6.0_dp + 12.0_dp*rnd())
w = sqrt(k*k + m*m)
! the deficit 1 - v = m^2 / (w (w + k)), computed without
cancellation
if (m*m/(w*(w + k)) > 0.0_dp) nslow = nslow + 1
end do
write(*, '(a,es9.2)') ' massless group velocity, max |v/c - 1| over
2000 frequencies: ', dmax
call check('every massless mode moves at c, independent of
frequency (to 1e-9)', dmax < 1.0e-9_dp)
call check('every massive mode moves below c (2000 of 2000)', nslow
== 2000)
mg = 1.0e-18_dp; ev = 1.0e-6_dp
dv = 0.5_dp*(mg/ev)**2
write(*, '(a,es9.2)') ' a photon mass at the bound 1e-18 eV slows a
1 microeV radio photon by dv/c = ', dv
call check('the photon-mass bound leaves light
frequency-independent below 1e-24 even at radio', dv < 1.0e-24_dp)
dv = 1.74_dp/(130.0e6_dp*3.15576e7_dp)
write(*, '(a,es9.2)') ' GW170817: 1.74 s after 130 million years of
flight, dv/c <= ', dv
call check('gravity and light, both massless, arrive together to
below 1e-15', dv < 1.0e-15_dp)
end subroutine block_b

subroutine block_c()
integer :: w, s, n
write(*, '(a)') ' C · helicity: parity reverses it and keeps the
frequency; the energy never reads it'
n = 0
do w = -500, 500
do s = -1, 1, 2
if (-s /= s .and. -(-s) == s) n = n + 1
end do
end do
call check('parity is an involution, moves every helicity, and
keeps the frequency (2002 states)', n == 2002)
end subroutine block_c

subroutine block_d()
integer :: n, cp, nok
write(*, '(a)') ' D · C-parity (-1)^n of n photons: Furry's
selection and positronium'
nok = 0; cp = 1
do n = 0, 40
if (n > 0) cp = -cp
if ((cp == 1) .eqv. (mod(n, 2) == 0)) nok = nok + 1
end do
call check('C-parity of 0 to 40 photons is +1 exactly on even
counts (41 of 41)', nok == 41)
call check('parapositronium (C = +1) decays to two photons,
orthopositronium (C = -1) to three', &
(-1)**2 == 1 .and. (-1)**3 == -1)
end subroutine block_d

subroutine block_e()
integer :: i, nf, nk
real(dp) :: pa, pb, g, v, d, worst, a, b, c, ip, im
write(*, '(a)') ' E · detection is registration: which-path and
fringe trade exactly, D^2 + V^2 = 1'
worst = 0.0_dp
do i = 1, 20000
pa = rnd(); pb = 1.0_dp - pa; g = rnd()
v = 2.0_dp*sqrt(pa*pb)*g
d = sqrt((pa - pb)**2 + 4.0_dp*pa*pb*(1.0_dp - g*g))
worst = max(worst, abs(d*d + v*v - 1.0_dp))
end do
write(*, '(a,es9.2)') ' max |D^2 + V^2 - 1| over 20000 pure
two-path states: ', worst
call check('Englert's duality holds with equality on every pure
state (to 1e-14)', worst < 1.0e-14_dp)
nf = 0; nk = 0
do i = 1, 20000
a = real(int(10.0_dp*rnd()) - 5, dp); b = real(int(10.0_dp*rnd())
- 5, dp)
c = real(int(3.0_dp*rnd()) - 1, dp)
ip = a*a + b*b + 2.0_dp*a*b*c; im = a*a + b*b - 2.0_dp*a*b*c
if ((ip /= im) .eqv. (a /= 0.0_dp .and. b /= 0.0_dp .and. c /=
0.0_dp)) nf = nf + 1
if (c == 0.0_dp .and. ip == im) nk = nk + 1
end do
call check('a fringe exists exactly when both paths are open and
the side is forgotten (20000)', nf == 20000)
call check('registering the side erases the fringe on every
sample', nk > 0)
end subroutine block_e

subroutine block_f()
integer :: n, i, j, nbad
real(dp) :: p, s1, s2, g2c, g2t, x(8), m1, m2, ga, sg
```

```

write(*,'(a)') 'F · the quantum is the field's: g2 = 0 for one
photon, >= 1 for every classical field'
! coherent (Poisson, mean 1) and thermal (geometric, mean 1)
photon-number statistics
s1 = 0.0_dp; s2 = 0.0_dp; p = exp(-1.0_dp)
do n = 0, 120
  if (n > 0) p = p/real(n, dp)
  s1 = s1 + real(n, dp)*p; s2 = s2 + real(n*(n - 1), dp)*p
end do
g2c = s2/(s1*s1)
s1 = 0.0_dp; s2 = 0.0_dp
do n = 0, 1000
  p = 0.5_dp**(n + 1)
  s1 = s1 + real(n, dp)*p; s2 = s2 + real(n, dp)*real(n - 1, dp)*p
end do
g2t = s2/(s1*s1)
write(*,'(a,f9.6,a,f9.6,a)') ' one photon g2 = 0; coherent g2 =
', g2c, ' ; thermal g2 = ', g2t, ' '
call check('one photon: g2 = 0; coherent light: g2 = 1; thermal
light: g2 = 2', &
  abs(g2c - 1.0_dp) < 1.0e-12_dp .and. abs(g2t - 2.0_dp) <
  1.0e-12_dp .and. 1*(1 - 1) == 0)

nbad = 0
do i = 1, 10000
  m1 = 0.0_dp; m2 = 0.0_dp
  do j = 1, 8
    x(j) = rnd()*3
    m1 = m1 + x(j); m2 = m2 + x(j)*x(j)
  end do

```

A · light is the line: every frequency, one side; the frequency is the height
 PASS 2001 photon frequencies: all on the line under both carriers, height = frequency
 the spectrum from radio to gamma, 1.00E-09 eV to 1.00E+11 eV: one side, 201 heights
 PASS twenty decades of photon energy occupy one side of the seat
 B · the speed hierarchy: massless modes at c at every frequency; massive modes below
 massless group velocity, max $|v/c - 1|$ over 2000 frequencies: 1.87E-10
 PASS every massless mode moves at c, independent of frequency (to 1e-9)
 PASS every massive mode moves below c (2000 of 2000)
 a photon mass at the bound 1e-18 eV slows a 1 microeV radio photon by $dv/c = 5.00E-25$
 PASS the photon-mass bound leaves light frequency-independent below 1e-24 even at radio
 GW170817: 1.74 s after 130 million years of flight, $dv/c \leq 4.24E-16$
 PASS gravity and light, both massless, arrive together to below 1e-15
 C · helicity: parity reverses it and keeps the frequency; the energy never reads it
 PASS parity is an involution, moves every helicity, and keeps the frequency (2002 states)
 D · C-parity $(-1)^n$ of n photons: Furry's selection and positronium
 PASS C-parity of 0 to 40 photons is +1 exactly on even counts (41 of 41)

```

  m1 = m1/8.0_dp; m2 = m2/8.0_dp
  if (m2 < m1*m1*(1.0_dp - 1.0e-12_dp)) nbad = nbad + 1
end do
call check('10000 classical intensity distributions: every one has
g2 >= 1', nbad == 0)
ga = 0.18_dp; sg = 0.06_dp
write(*,'(a,f5.2,a,f5.1,a)') ' Grangier, Roger and Aspect:
anticorrelation ', ga, ' , below the classical 1 by ', &
(1.0_dp - ga)/sg, ' sigma'
call check('the measured single-photon anticorrelation is below
every classical field by > 10 sigma', &
(1.0_dp - ga)/sg > 10.0_dp)
end subroutine block_f

subroutine block_g()
  real(dp) :: floor, ephot
  write(*,'(a)') 'G · the price of a detection: each registration
commits at least one bit'
  floor = KB*300.0_dp*log(2.0_dp)
  ephot = HP*CL/550.0e-9_dp
  write(*,'(a,e12.5,a,f7.1,a)') ' k_B T ln 2 at 300 K = ', floor, '
J; a 550 nm photon carries ', &
  ephot/floor, ' floors'
  call check('the floor is 2.870978885e-21 J and a visible photon
pays its own registration', &
  abs(floor/2.8709788850787237e-21_dp - 1.0_dp) <
  1.0e-14_dp .and. ephot > floor)
end subroutine block_g
end program light_twin

```

PASS parapositronium (C = +1) decays to two photons, orthopositronium (C = -1) to three
 E · detection is registration: which-path and fringe trade exactly, $D^2 + V^2 = 1$
 max $|D^2 + V^2 - 1|$ over 20000 pure two-path states: 4.44E-16
 PASS Englert's duality holds with equality on every pure state (to 1e-14)
 PASS a fringe exists exactly when both paths are open and the side is forgotten (20000)
 PASS registering the side erases the fringe on every sample
 F · the quantum is the field's: g2 = 0 for one photon, >= 1 for every classical field
 one photon g2 = 0; coherent g2 = 1.000000; thermal g2 = 2.000000
 PASS one photon: g2 = 0; coherent light: g2 = 1; thermal light: g2 = 2
 PASS 10000 classical intensity distributions: every one has g2 >= 1
 Grangier, Roger and Aspect: anticorrelation 0.18, below the classical 1 by 13.7 sigma
 PASS the measured single-photon anticorrelation is below every classical field by > 10 sigma
 G · the price of a detection: each registration commits at least one bit
 k_B T ln 2 at 300 K = 2.87098E-21 J; a 550 nm photon carries 125.8 floors
 PASS the floor is 2.870978885e-21 J and a visible photon pays its own registration
 BATTERY-JSON: {"checks":16,"failures":0}

APPENDIX C: THE SEAT AND THE JUDGMENT

```

PhysOST 1.0.5p · RECEIPT · mode quick
TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caae842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
MANIFEST sha256 a4108cd26bcbe6ef, the files that ran
GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 PhysOS Proofs read from II.10, record
lines complete · source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6
retirements string-scanned, 3 held by review, clean
EXITS ground 0 · registerA RA_TOE_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB
One_Cut_Resolution 0 · registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure
0 · registerB Closure_Executed 0 · registerB Armed_Seal 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB
RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin
0
REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}
WITNESS 5 of 5 twins ran, each held to one battery line with zero failures
CHAIN · D0 83f872725ad9 -> D1 90fc702bbd1 -> D2 492e519daab9 -> D3 6618bec5164b
ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)
SEAT EARNED · this run

```

```

SCREEN forbidden: none | imports: 0 | axioms declared: 0
laws: 63
JUDGMENT refused 63 of 63 as proof failures | survived none | parse failures none | planted vacuous law survives its negation: True

```